

PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO

Test Booklet Code

This Booklet contains 24 pages.

L1

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SUBJECT : PHYSICS

Topic : SYLLABUS-1.

- Which of the following pair represent the same dimensions :-
 (1) Watt and Newton-metre
 (2) Pascal and Joule
 (3) Newton and Joule
 (4) Joule and Newton-metre
- If x is length, $v = \frac{dx}{dt}$ is velocity and $a = \frac{dv}{dt}$ is acceleration, then which option is not possible :-
 (t = time)
 (1) $\left(\frac{dv}{dt} + \frac{d^2x}{dt^2}\right)$
 (2) $\left(\frac{dv}{dt} - \frac{d^2x}{dt^2}\right)$
 (3) $\left(\frac{d^2x}{dt^2} + x \frac{dv}{dt}\right)$
 (4) $\left(v \frac{dv}{dx} + 5 \frac{d^2x}{dt^2}\right)$
- In a relation $y = A \sin(e^{\alpha x} + \beta)$ here y and x represent length then dimensions of $\frac{A}{\alpha\beta}$ will be :-
 (1) $[ML^{-1}T^0]$ (2) $[M^0L^2T^0]$
 (3) $[M^0L^{-2}T^0]$ (4) $[M^0L^2T]$
- If force, $F = \rho^a A^b v^c$, here ρ is density, A is area and v is velocity then value of a, b, c will be :-
 (1) $a = 1, b = 1, c = 2$
 (2) $a = 2, b = 1, c = 1$
 (3) $a = 1, b = 2, c = 1$
 (4) $a = 1, b = 2, c = 2$
- If acceleration $a_t = m^a v^b t^c$ where m is mass, v is velocity and t is time then value of $(a^2 + bc)$ will be :-
 (1) Zero (2) -1 (3) 2 (4) -2
- Two resistance $R_1 = 1000 \pm 5\% \Omega$ and $R_2 = 200 \pm 5\% \Omega$ are connected in series. The percentage error in equivalent resistance will be :-
 (1) 10% (2) 25%
 (3) 15% (4) 5%
- A physical quantity represented as $x = \frac{A^2 B^{1/3}}{\sqrt{CD}}$, if percentage error in A is 2%, B is 3%, C is 4% and D is 6%, then percentage error in x will be :-
 (1) 5% (2) 15%
 (3) 10% (4) 12%
- The time period of simple pendulum is given by $T = 2\pi\sqrt{\frac{\ell}{g}}$. Measured value of ℓ is 40.0 cm and time for 200 oscillations is found to be 400 s by using of resolution 1s. Maximum percentage error in g will be if it is measured from above relation for time period :-
 (1) 1.25% (2) 0.75%
 (3) 2.50% (4) 0.5%
- The radius of sphere is measured to be (5.00 ± 0.05) cm. If percentage error in its volume and surface area are $x\%$ and $y\%$ then $|x| + |y|$ will be :-
 (1) 2 (2) 5 (3) 7 (4) 3
- If 30 litres of water per minute is lifted from a well through 10 m depth and delivered with a velocity of $2\sqrt{15}$ m/s then the power of the motor is :- (take $g = 10 \text{ m/s}^2$ and $\rho_{\text{water}} = 10^3 \text{ kg/m}^3$)
 (1) 50 watt
 (2) 80 watt
 (3) 65 watt
 (4) 72 watt

PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO

11. A block of mass 3 kg is initially at rest on a horizontal frictionless surface. A horizontal force $\vec{F} = (x^2 + x)\hat{i}$ N acts on it, when the block is at $x = 0$. The maximum kinetic energy of the block between $x = 0$ to $x = 1$ m is :-

- (1) 1.2 J (2) 0.83 J
(3) 0.72 J (4) 0.98 J

12. Consider the following two statements :-

(P) Linear momentum of a system of particles is zero.

(Q) Kinetic energy of a system of particles is zero.

Then

- (1) P does not imply Q and Q does not imply P.
(2) P implies Q but Q does not imply P.
(3) P does not imply Q but Q implies P.
(4) P implies Q and Q implies P.

13. Match the following column-

	Column-1		Column-2
(a)	Erg	(p)	1.6×10^{-19} J
(b)	Electron volt (eV)	(q)	3.6×10^6 J
(c)	Calorie (cal)	(r)	10^{-7} J
(d)	Kilowatt hour (kWh)	(s)	4.186 J

- (1) (a-r), (b-p), (c-s), (d-q)
(2) (a-p), (b-r), (c-q), (d-s)
(3) (a-p), (b-q), (c-r), (d-s)
(4) (a-s), (b-r), (c-q), (d-p)

14. A uniform chain has a mass m and length ℓ . It is placed on frictionless table with a length $\ell/4$ hanging over the edge. The chain begins to slide down. What is the velocity v with which the end slides away from the edge?

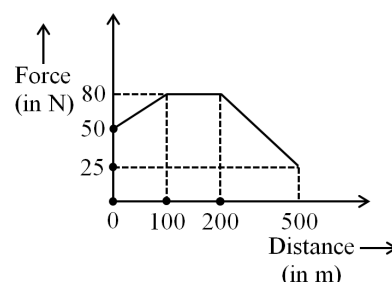
- (1) $\sqrt{g\ell}$ (2) $\sqrt{\frac{15}{32}g\ell}$
(3) $\sqrt{\frac{8}{16}g\ell}$ (4) $\sqrt{\frac{15}{16}g\ell}$

15. **Assertion (A)** : A spring does negative work only when it is compressed.

Reason (R) : The spring force always act in the opposite direction of deformation in the spring.

- (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

16. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle becomes $(\alpha \times 10^3)$ J after it has travelled first 200 m distance, then value of α is-

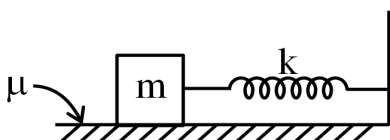


- (1) 1.45 (2) 145
(3) 14.5 (4) 14500
17. A ball of mass 800 g is thrown vertically upwards with initial velocity 20 m/s. If due to air resistance and gravity ball reached maximum height of 16 m. The work done by air resistance is- ($g = 10 \text{ m/s}^2$)
- (1) 32 J
(2) 160 J
(3) - 32 J
(4) -160 J

18. A particle in a certain conservative force field has potential energy given by $U = 2xyz^{-1}$. The force exerted on it at position $(1, 0, -1)$ is-

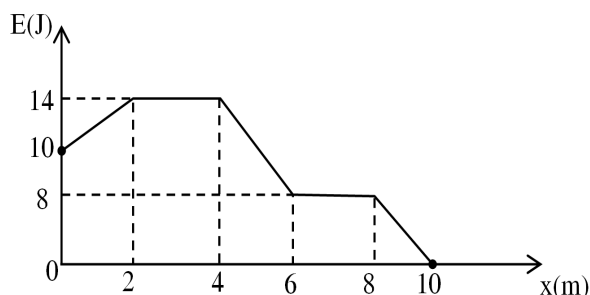
- (1) $-\hat{i} + 2\hat{j}$
- (2) $2\hat{j}$
- (3) $-2\hat{i}$
- (4) $-2\hat{k} + 2\hat{j}$

19. Find the maximum energy stored in the spring shown in the figure, for which the block remains stationary on a rough horizontal surface.



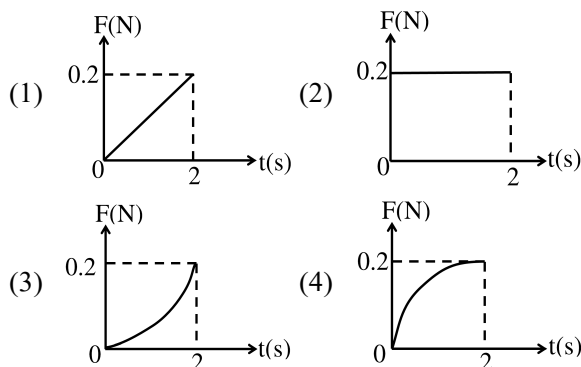
- (1) $\frac{\mu mg}{k}$
- (2) $\frac{(\mu mg)^2}{k}$
- (3) $\frac{\mu(mg)^2}{k}$
- (4) $\frac{(\mu mg)^2}{2k}$

20. The kinetic energy (E) of a particle moving along x -axis varies with position (x) as shown in figure. The magnitude of force acting on the particle at $x = 5\text{m}$.

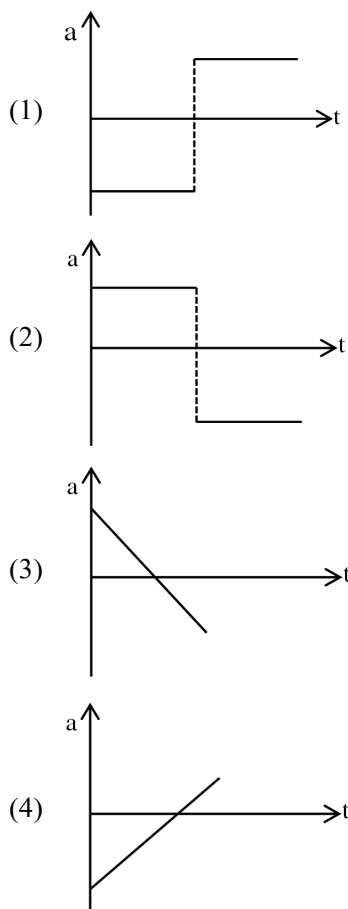
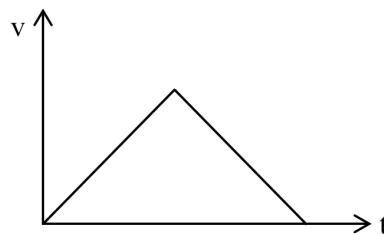


- (1) 3 N
- (2) 5 N
- (3) 12 N
- (4) 8 N

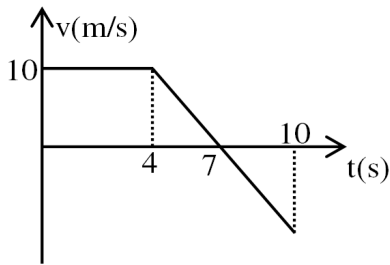
21. If kinetic energy of a particle of mass 10g moving along a straight line under the influence of a force varies with time as $K = (2t^2)$ J. The force acting on the particle varies with time as-



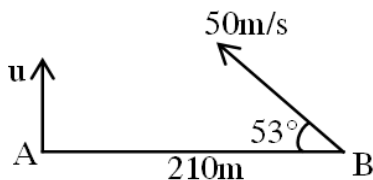
22. Choose correct graph of a - t for given v - t graph :



23. Find out distance covered by particle in 10s in given graph -



- (1) 40 m (2) 110 m
(3) 70 m (4) 90 m
24. Speed of boat in still water is 5 km/h and river is flowing with speed 3 km/h. Find out time taken by boat to move 8 km in down stream and come back.
- (1) 1 hr
(2) 4 hr
(3) 5 hr
(4) 2 hr
25. Car A is 100 m towards North of car B. Car A starts moving towards east with speed 10 m/s simultaneously car B starts to move towards north with same speed find out time when separation between the cars become minimum.
- (1) 5 s (2) $\frac{5}{\sqrt{2}}$ s
(3) $5\sqrt{2}$ s (4) $\frac{5}{2}$ s
26. Find out projection speed of particle A such that A can collide with particle B in given figure.



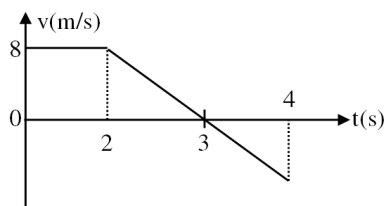
- (1) 40 m/s (2) 30 m/s
(3) 50 m/s (4) 20 m/s

27. A particle is projected with speed 50 m/s at an angle 53° from vertical. Choose the correct option -

(A)	Max. height (m)	(P)	6
(B)	Range (m)	(Q)	120
(C)	Time of flight (s)	(R)	240
		(S)	45

- (1) A-Q, B-R, C-S
(2) A-S, B-Q, C-R
(3) A-S, B-R, C-P
(4) A-R, B-S, C-Q
28. If equation of trajectory of projectile is $y = x - \frac{x^2}{80}$ than find out maximum height of this projectile.
- (1) 20 m (2) 40 m
(3) 60 m (4) 80 m
29. If position of a particle with time is $x = 2t - t^2$ than find out distance covered by it in first 2 s.
- (1) Zero (2) 1 m (3) 2 m (4) 4 m
30. A particle is moving on a circular path of radius 2m. Find out its displacement when it covers angle $\frac{2\pi}{3}$ radian.
- (1) $2\sqrt{3}$ m (2) $\frac{2}{\sqrt{3}}$ m
(3) 2m (4) $\sqrt{3}$ m
31. A particle has initial velocity $(8\hat{i} + 6\hat{j})$ and has acceleration $(0.3\hat{i} + 0.4\hat{j})$. It's speed after 10s is :-
- (1) $7\sqrt{2}$ m/s
(2) $11\sqrt{2}$ m/s
(3) 15 m/s
(4) $\sqrt{221}$ m/s

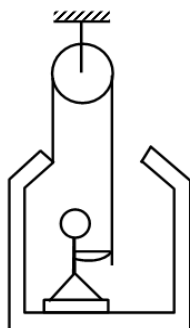
32. Find out average speed of particle in 4s in given graph



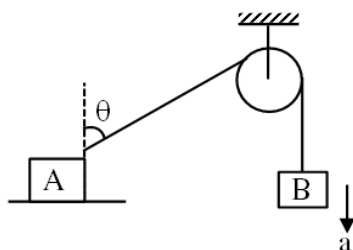
- (1) 6 m/s (2) 7 m/s (3) 8 m/s (4) 10 m/s
33. Car A is moving towards North with speed 5 m/s. Its driver observes that car B is moving towards East. If car B is moving towards 30° N of E. Find out velocity of car B with respect to car A.

- (1) 5 m/s (2) $5\sqrt{3}$ m/s
(3) $\frac{5}{\sqrt{3}}$ m/s (4) 10 m/s

34. A man of 60 kg is standing on a weighing machine in a cage of 30 kg in the given figure. If the system is in equilibrium then find out reading of weighing machine :



- (1) 100 N (2) 150 N (3) 300 N (4) 600 N
35. Find out acceleration of block A if block B is moving downward with acceleration 'a' in given figure :



- (1) $a \cos \theta$ (2) $a \sin \theta$
(3) $a \sec \theta$ (4) $a \operatorname{cosec} \theta$

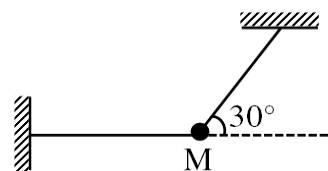
36. While walking on ice one should take small steps to avoid slipping. This is because small steps ensure –

- (1) Large friction
(2) Smaller friction
(3) Larger normal force
(4) Smaller normal force

37. N bullets each of mass 'm' kg are fired with a velocity $v \text{ ms}^{-1}$ at the rate of 'n' bullets per second upon a wall. Bullets get embedded into the wall. The reaction offered by the wall to the bullets is given by :

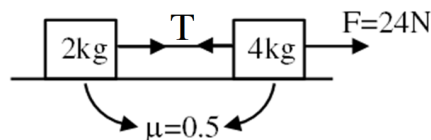
- (1) nmv
(2) $\frac{Nm v}{n}$
(3) $n \frac{Nm}{v}$
(4) $n \frac{N v}{m}$

38. A mass 'M' is hung with a light inextensible string. Tension in horizontal part of string is:



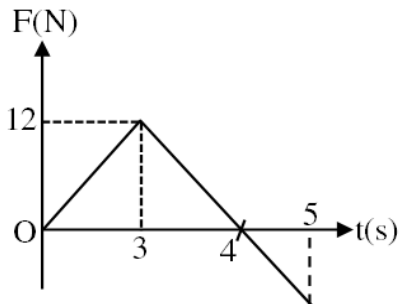
- (1) $\sqrt{3}Mg$
(2) $\sqrt{2}Mg$
(3) $\frac{Mg}{\sqrt{3}}$
(4) $\frac{Mg}{2}$

39. Find out tension in the string in given figure :

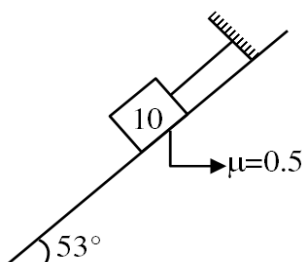


- (1) 4 N (2) 10 N (3) 24 N (4) 30 N

40. Force-time graph of a particle is given. Find out impulse in 5 seconds.

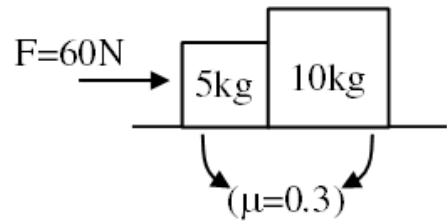


- (1) 18 N-s
(2) 30 N-s
(3) 9 N-s
(4) 6 N-s
41. A balloon has 2 gm of air. A small hole is projected into it. The air comes out with a velocity of 4 m/s. If the balloon shrinks completely in 2.5 s, the average force acting on the balloon is :
- (1) 0.008 N
(2) 0.0032 N
(3) 8 N
(4) 3.2 N
42. Find out tension in the string in given figure :

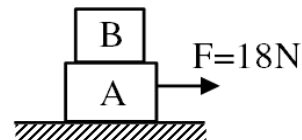


- (1) Zero
(2) 80 N
(3) 50 N
(4) 30 N

43. Find out normal contact force between 5 kg and 10 kg in given figure :



- (1) 20 N
(2) 30 N
(3) 40 N
(4) 60 N
44. A block of mass 2 kg is placed on the floor. The coefficient of static friction is 0.4. If a force of 2.8 N is applied on the block parallel to the floor, the force of friction between the block and the floor (taking $g = 10 \text{ m/s}^2$) is :
- (1) 2.8 N
(2) 8 N
(3) 2 N
(4) Zero
45. A block B of mass 2 kg is placed on another block A of mass 4 kg which rests on a smooth horizontal surface. If $\mu = 0.5$ between A and B and a force of 18 N is applied on A, the friction between the blocks :



- (1) 10 N
(2) 8 N
(3) 6 N
(4) 4 N

SUBJECT : CHEMISTRY

Topic : SYLLABUS-1.

46. What will be the concentration of acetate ion in a 1.0 M solution of acetic acid containing 0.1 M HCl [$K_a = 2 \times 10^{-5}$] :-
 (1) 2×10^{-3} M (2) 2×10^{-4} M
 (3) 2×10^{-2} M (4) 2×10^{-6} M
47. For which reaction will $K_p = K_c$?
 (1) $2\text{CO(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO}_2\text{(g)}$
 (2) $\text{H}_2\text{(g)} + \text{Br}_2\text{(g)} \rightleftharpoons 2\text{HBr(g)}$
 (3) $\text{C(s)} + 2\text{H}_2\text{O(g)} \rightleftharpoons 2\text{H}_2\text{(g)} + \text{CO}_2\text{(g)}$
 (4) $\text{PCl}_5\text{(g)} \rightleftharpoons \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$
48. On doubling the volume of the vessel, which of the following equilibrium will be shifted to right ?
 (1) $\text{PCl}_5\text{(g)} \rightleftharpoons \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$
 (2) $\text{H}_2\text{(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons 2\text{HCl(g)}$
 (3) $2\text{CO(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO}_2\text{(g)}$
 (4) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$
49. On adding inert gas to the equilibrium $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ at constant pressure, the degree of dissociation will :-
 (1) Not change (2) Decrease
 (3) Increase (4) None of these
50. The formula used to evaluate the pH of monobasic weak acid is :-
 (1) $\text{pH} = \text{C}\alpha$
 (2) $\text{pH} = 1/2 \text{pK}_a - 1/2 \log C$
 (3) $\text{pH} = \sqrt{K_a C}$
 (4) $\text{pH} = \text{pK}_a + \log C$
51. What is the pH of a weak acid HA having 0.004 M concentration and $\text{pK}_a = 5$?
 (1) 4.70 (2) 3.70
 (3) 7.40 (4) 5
52. If $\text{pOH} \geq 5$, then possible nature of aqueous Solution is :-
 (1) acidic
 (2) basic
 (3) neutral
 (4) any one is possible
53. Which of the following will make acidic buffer ?
 (1) 50 mL of 0.1M NaOH + 25 mL of 0.1M CH_3COOH
 (2) 100 mL of 0.1M HCl + 100 mL of 0.1M NaOH
 (3) 100 mL of 0.1M HCl + 200 mL of 0.1M NH_4OH
 (4) 25 mL of 0.1M NaOH + 50 mL of 0.1M CH_3COOH
54. 0.1M formic acid solution is titrated against 0.1M NaOH solution. What would be the difference in pH between 1/5 and 4/5 stages of neutralization of acid?
 (1) $2\log(3/4)$ (2) $2\log(1/5)$
 (3) $\log(1/3)$ (4) $2\log 4$
55. What is the pH at which 0.01 M Co^{+2} ions in solution begin to precipitate as Co(OH)_2 ? K_{sp} of $\text{Co(OH)}_2 = 25 \times 10^{-16}$:-
 (1) 6.3 (2) 7.3 (3) 10.8 (4) 7.7
56. Sodium nitrate, heated in the presence of an excess of H_2 , forms H_2O according to two step process:
 $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
 How many grams of NaNO_3 are required to form 9 g of H_2O ?
 (1) 21.3 (2) 42.5 (3) 69.0 (4) 85.0

57. When mixture of 6 L of methane and propane were burnt, 7 L CO_2 was obtained. Then volume of methane and propane in mixture will be :-
 (1) 5.5 L and 0.5 L (2) 6 L and 1 L
 (3) 3 L and 4 L (4) 5 L and 2 L
58. A 1000 gm sample of NaOH contains 3 moles of oxygen atoms. What is the % purity of NaOH sample :-
 (1) 14% (2) 100% (3) 12% (4) 24%
59. If 4 mole of Mn^{2+} will loose 16 mole of electrons, to form a single product then it may form:
 (1) MnO_2 (2) MnO_4^-
 (3) MnO_4^{2-} (4) Mn_2O_5
60. Which substance is serving as a reducing agent in the following reaction ?
 $14\text{H}^+ + \text{Cr}_2\text{O}_7^{2-} + 3\text{Ni} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{Ni}^{2+}$
 (1) H_2O (2) Ni
 (3) H^+ (4) $\text{Cr}_2\text{O}_7^{2-}$
61. How many mole of electrons are needed to convert one mole of nitrate ion to hydrazine (N_2H_4) ?
 (1) 8 (2) 7 (3) 5 (4) 3
62. Reaction (A) $\text{S}^{2-} + 4\text{H}_2\text{O}_2 \rightarrow \text{SO}_4^{2-} + 4\text{H}_2\text{O}$
 (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 The true statement regarding the above reactions is :
 (1) H_2O_2 acts as reductant in both the reactions.
 (2) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).
 (3) H_2O_2 acts as an oxidant in both the reactions.
 (4) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction(B)
63. In which of the following, entropy decreases ?
 (1) Dissolving sucrose in water
 (2) Sublimation of camphor
 (3) Melting of ice
 (4) Stretching of rubber band
64. The enthalpy change for transition of liquid water to steam is 3.73 kJ mol^{-1} at 373 K. Find ΔS in J/mol K :-
 (1) 100 J/mol K (2) 10 J/mol K
 (3) 1000 J/mol K (4) None of these
65. A gas is allowed to expand from a volume of 1.0 L to 10.0 L against a constant external pressure of 0.50 atm. If the gas absorbs 250 J of heat from the surroundings, what are the values of q, w and ΔE ?
- | | q | w | ΔE |
|-----|--------|--------|------------|
| (1) | 250 J | -455 J | -205 J |
| (2) | -250 J | -455 J | -710 J |
| (3) | 250 J | 455 J | 710 J |
| (4) | -250 J | 455 J | 205 J |
66. In a cyclic process, the amount of heat given to a system is equal to
 (1) Net work done by the system
 (2) Net increase in internal energy
 (3) Net decrease in internal energy
 (4) Net change in specific heat
67. Enthalpies of formation of CO(g) , $\text{CO}_2\text{(g)}$, $\text{N}_2\text{O(g)}$ and $\text{N}_2\text{O}_4\text{(g)}$ are -110, -393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of ΔH for the reaction :
 $\text{N}_2\text{O}_4\text{(g)} + 3\text{CO(g)} \rightarrow \text{N}_2\text{O(g)} + 3\text{CO}_2\text{(g)}$
 (1) $+778 \text{ kJ mol}^{-1}$ (2) $-211.7 \text{ kJ mol}^{-1}$
 (3) $+211.7 \text{ kJ mol}^{-1}$ (4) -778 kJ mol^{-1}

68. For the hypothetical reaction
 $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$
 $\Delta_r G^\circ$ and $\Delta_r S^\circ$ are 20 kJ/mol and $-20 \text{ JK}^{-1} \text{ mol}^{-1}$ respectively at 200 K. If $\Delta_r C_p$ is $20 \text{ JK}^{-1} \text{ mol}^{-1}$ then $\Delta_r H^\circ$ at 400 K is :-
 (1) 20 kJ/mol (2) 7.98 kJ/mol
 (3) 28 kJ/mol (4) None of these
69. In order to decompose 9 gm water into H_2 and O_2 , 142.5 kJ heat is required. Hence the enthalpy of formation of water is :-
 (1) +285 kJ (2) -285 kJ
 (3) +142.5 kJ (4) -142.5 kJ
70. The heat of combustion of ethylene $C_2H_4(g)$ is -1420 kJ/mole. The number of litres of C_2H_4 at NTP that would evolve 355 kJ on combustion is
 (1) 2.8 (2) 8.4 (3) 5.6 (4) 11.2
71. 10 mol ammonium carbamate is heated in a closed container so that at equilibrium, 12 atm total pressure was obtained, what will be K_p for the following reaction ?
 $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
 (1) 512 (2) 256 (3) 144 (4) 1
72. Which is correct statement regarding CrO_5
 A : Oxidation number of Cr is +6.
 B : Four oxygen-atoms of molecule are involved in peroxide linkage.
 C : Only one oxygen-atom has O.N. = -2
 (1) All are correct (2) B, C are correct
 (3) A, C are correct (4) Only B is correct
73. In the reaction ;
 $As_2S_3 + HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$,
 the element oxidised is/are :-
 (1) As only (2) S only
 (3) N only (4) As and S both
74. If M_4O_5 compound is written as $M_x(+1) M_y(+3)O_5$ then calculate the value of x & y:-
 (1) 1, 5 (2) 2, 2 (3) 3, 1 (4) 1, 3
75. How many moles of hydrazine (N_2H_4) can be oxidized into N_2 using $2/3$ mole of bromate ions ($BrO_3^- \rightarrow Br^-$) :-
 (1) $\frac{1}{3}$ (2) 1 (3) 1.5 (4) $\frac{2}{3}$
76. Calculate the work done when 1 mol of an ideal gas is compressed reversibly from 1 bar to 4 bar at a constant temperature of 300 K -
 (1) 3.45 kJ (2) -8.02 kJ
 (3) 18.02 kJ (4) -14.01 kJ
77. The bond dissociation energies for Cl_2 , I_2 and ICl are 242.3, 151 and 211.3 kJ/mol respectively. The enthalpy of sublimation of iodine is 62.8 kJ/mol. What is the standard enthalpy of formation of $ICl(g)$?
 (1) -211.3 kJ/mol (2) -14.6 kJ/mol
 (3) 16.8 kJ/mol (4) 33.5 kJ/mol
78. The sublimation enthalpy of $I_2(s)$ is 57.3 kJ/mol and the enthalpy of fusion is 15.5 kJ/mol. The enthalpy of vaporisation of I_2 is-
 (1) 41.8 kJ/mol (2) -41.8 kJ/mol
 (3) 72.8 kJ/mol (4) -72.8 kJ/mol
79. What is the pH of a solution containing 25.0 mL each of 0.1 M NaOH and 0.08 M HCl and final solution is diluted upto 500 mL ?
 (1) 3 (2) 11
 (3) 12 (4) 13
80. $x A_{(s)} \rightleftharpoons y B_{(g)} + z C_{(g)}$
 If $\frac{K_c}{K_p} = (RT)^{-2}$, then which is correct -
 (1) $y + z - x = -2$ (2) $y + z - x = 2$
 (3) $y + z = -2$ (4) $y + z = 2$

81. The number of oxygen atoms in 0.2 mole of sodium carbonate decahydrate is :-
 (1) 1.56×10^{24} (2) 1.56×10^{23}
 (3) 1.56×10^{25} (4) 3.12×10^{24}
82. How many moles of P_4 can be produced by reaction of 0.10 moles $Ca_5(PO_4)_3F$, 0.36 moles SiO_2 and 0.90 moles C according to the following reaction ?
 $4Ca_5(PO_4)_3F + 18SiO_2 + 30C \rightarrow 3P_4 + 2CaF_2 + 18CaSiO_3 + 30CO$
 (1) 0.060 (2) 0.030
 (3) 0.045 (4) 0.075
83. Calculate ΔG° (kJ/mol) at $127^\circ C$ for a reaction with $K_{equilibrium} = 10^5$
 (1) -38.294 (2) -16.628
 (3) -9.16 (4) None of these
84. When CO_2 dissolves in water, the following equilibrium is established
 $CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$
 for which the equilibrium constant is 3.8×10^{-7} and $pH = 6.0$. The ratio of $[HCO_3^-]$ to $[CO_2]$ would be :-
 (1) 3.8×10^{-13}
 (2) 3.8×10^{-1}
 (3) 6.0
 (4) 13.4
85. The bond energies of $C - C$, $C = C$, $H - H$ and $C - H$ bonds are 350, 600, 400 and 410 kJ per mole respectively. The heat of hydrogenation of ethylene (C_2H_4) is :-
 (1) -170 kJ mol^{-1}
 (2) -260 kJ mol^{-1}
 (3) -400 kJ mol^{-1}
 (4) -450 kJ mol^{-1}
86. 0.01 M $HA(aq.)$ is 2% ionized, $[OH^-]$ of solution is :-
 (1) 2×10^{-4} (2) 10^{-8}
 (3) 5×10^{-11} (4) 5×10^{-12}
87. If 1 mole of H_3PO_x is completely neutralized by 80 gm of $NaOH$, select the correct statement :-
 (1) $x = 2$ and acid is monobasic
 (2) $x = 3$ and acid is dibasic
 (3) $x = 4$ and acid is tribasic
 (4) All are correct
88. What is the pH of a solution prepared by mixing 20 ml. of 0.250 M sodium formate with 30 mL of 0.100 M HCl (pK_a for $HCOOH = 3.75$) :-
 (1) 3.57 (2) 4.18
 (3) 5.22 (4) None
89. Consider the following data for series of hydrogen halide Bronsted acids :-
- | Acid | HF | HCl | HBr | HI |
|-------|----------------------|-----------------|-----------------|--------------------|
| K_a | 7.2×10^{-4} | 1×10^6 | 1×10^9 | 3×10^{11} |
- Which of these Bronsted acids would have the weakest conjugate base ?
 (1) HF (2) HCl
 (3) HBr (4) HI
90. Heat of combustion of carbon in diamond and amorphous form are -94.3 and -97.6 kcal/mole. The heat required to convert 18 gm carbon from diamond to amorphous form is:-
 (1) -4.95 kcal
 (2) $+4.95$ kcal
 (3) -3.3 kcal
 (4) $+3.3$ kcal

SUBJECT : BIOLOGY

Topic : SYLLABUS-1.

91. Two organisms belong to same class but belong to different families, they may be kept in same :-
 (1) Genus (2) Species
 (3) Order (4) Variety
92. Which of the following is not a correct statement?
 (1) The number of species that are known and described range between 1.7 to 1.8 billion.
 (2) The number and types of organisms present on earth referred as biodiversity.
 (3) As we explore new areas, and even old ones, new organisms are continuously being identified.
 (4) Each different kind of plant, animal or organism that you see, represent a species.
93. **Statement-I** : All living organisms can be classified into different taxa.
Statement-II : The word systematics is derived from the Latin word 'Systema'.
 (1) Both statement-I and statement-II are correct.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect but statement-II is correct.
 (4) Both statement-I and statement-II are incorrect.
94. **Statement-I** : Each scientific name has two components.
Statement-II : The first word in a biological name represents the specific epithet while second word denotes the genus.
 (1) Both statement I and II are correct
 (2) Both statement I and II are incorrect
 (3) Statement I is correct but statement II is incorrect.
 (4) Statement I is incorrect but statement II is correct
95. **Assertion (A)** : In natural classification systems, more weightage will be given to reproductive/sexual characters.
Reason (R) : Vegetative characters are more easily affected by environment as compared to sexual characters.
 (1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
 (2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.
 (3) **Assertion** is correct but **Reason** is incorrect.
 (4) **Assertion** is incorrect but **Reason** is correct.
96. Which of the following statements are true?
 (a) Classification given by Bentham and Hooker was of non-flowering plants, algae and bryophytes.
 (b) Numbers and codes are assigned to floral characters only, in numerical taxonomy.
 (c) Phylogenetic classification systems are based on evolutionary relationships between various organisms.
 (d) The earliest systems of scientific classification used only gross superficial morphological characters.
 (1) a and b only (2) c and d only
 (3) a, c and d only (4) a, b and c only
97. **Assertion (A)** - The problem of classification becomes more complex at higher taxonomic categories.
Reason (R) - As we go from species to Kingdom, the number of common characteristics goes on decreasing.
 (1) Both (A) and (R) are correct and (R) is correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not correct explanation of (A)
 (3) (A) is correct but (R) is incorrect
 (4) (A) is incorrect but (R) is correct

98. Given organisms belongs to how many families ?
Wheat, Mango, Brinjal, Potato, Datura, Lion, Cat, Dog

(1) Seven (2) Six (3) Five (4) Four

99. Find out the family and order of *Triticum aestivum* :-

	Family	Order
1	Anacardiaceae	Poales
2	Poaceae	Sapindales
3	Anacardiaceae	Sapindales
4	Poaceae	Poales

100. Monkey, Gorilla, Tiger and Cat belongs to different order but based on common features they all belongs to the same :

(1) Species
(2) Genus
(3) Family
(4) Class

101. Study the following statements :

I. Growth and Reproduction are not the defining character of living organism while metabolism, cellular organisation and consciousness are defining characters.

II. In multicellular organisms, reproduction refer to the production of progeny possessing features more or less similar to those of parents.

III. All higher animals have self consciousness.

IV. All organisms from the prokaryotes to the most complex eukaryotes can sense and respond to environmental cues.

Choose the correct option :-

(1) Only I, III, IV are correct
(2) Only I, II, IV are correct
(3) All are correct
(4) Only II and IV are correct

102. Read the following statements :

A. He was the earliest to attempt a more scientific basis for classification.

B. He use simple morphological characters to classify plants into trees, shrubs and herbs.

C. He also divided animals into two groups, those which had red blood and those that did not.

Above statements are related with -

(1) Aristotle (2) Linnaeus
(3) Theophrastus (4) R.H. Whittaker

103. Choose the incorrect statement with respect to three-domain system ?

(1) Carl Woese placed prokaryotes in two domains
(2) Carl Woese placed prokaryotes in two kingdoms
(3) Carl Woese placed eukaryotes in one domain
(4) Carl Woese placed eukaryotes in two kingdom

104. Which one was not the criteria used by R.H. Whittaker in five kingdom classification ?

(1) Thallus organisation
(2) Cell structure
(3) Growth habit
(4) Mode of nutrition

105. Who placed *Chlamydomonas* and *Chlorella* in kingdom Protista ?

(1) Aristotle
(2) Whittaker
(3) Linnaeus
(4) Bentham and Hooker

106. Sole members of kingdom Monera are -

(1) Bacteria
(2) Diatoms
(3) Desmids
(4) Lichens

107. Which one of the following is not the character of bacterial cell ?

- (1) Mesosomes (2) Genomic DNA
(3) 70 s ribosomes (4) Nuclear membrane

108. A special membranous structure formed by extension or infoldings or invagination of plasma membrane in to the cell is called as :-

- (1) Mesosomes (2) Ribosome
(3) Storage granules (4) Nucleoid

109. Which of the following structure help in DNA replication in bacterial cell ?

- (1) Lipopolysaccharide wall
(2) Peptidoglycan wall
(3) Mesosome
(4) Ribosome

110. Chromatophores are present in :-

- (1) *Albugo* (2) Slime moulds
(3) *Trypanosoma* (4) *Nostoc*

111. Match the column-I and II and select correct option :

Column-I		Column-II	
(I)	Halophiles	(p)	Hot springs
(II)	Thermoacidophiles	(q)	Marshy areas
(III)	Methanogens	(r)	Salty areas

- (1) I-p, II-q, III-r (2) I-q, II-p, III-r
(3) I-r, II-p, III-q (4) I-r, II-q, III-p

112. Select the correct statement from the following regarding protists :-

- (1) Members of protista are primarily terrestrial.
(2) All are multicellular eukaryotes
(3) Have well-defined nucleus
(4) No zygote formation.

113. Which of the following statements is correct for flagella of dinoflagellates ?

- (1) Only single flagellum lies in the transverse groove between the wall plates
(2) Only single flagellum lies in the longitudinal groove between the wall plates
(3) Two flagella, one lies longitudinally and the other transversely in a furrow between the wall plates.
(4) Flagella are absent

114. 'Diatomaceous earth' is used in :-

- (1) Polishing (2) Filtration of oils
(3) Filtration of syrups (4) All of the above

115. Which protista are photosynthetic in presence of sunlight but behave like heterotroph when deprived of sunlight by predated on other smaller organisms :-

- (1) Chrysophytes (2) Dinoflagellates
(3) Euglenoids (4) Slime moulds

116. Slime moulds are-

- (1) Photosynthetic protists
(2) Saprophytic protists
(3) Parasite protists
(4) Both (1) and (2)

117. Vegetative reproduction in fungi can take place by which process :-

- (1) Fragmentation (2) Fission
(3) Budding (4) All of the above

118. (A) Members of phycomycetes are found in aquatic habitat.

(B) Members of phycomycetes are obligate parasite on plants.

(C) The mycelium of phycomycetes is septate.

(D) In phycomycetes zoospores are produced exogenously.

(E) In phycomycetes, Asexual reproduction take place by non motile aplanospore.

Select the set of correct statements :

- (1) A, B and C (2) B, C and D
(3) C, D and E (4) A, B and E

119. Which of the following ascomycetes is the source of antibiotics ?

- (1) *Neurospora*
- (2) *Penicillium*
- (3) *Claviceps*
- (4) *Puccinia*

120. Match the following and choose correct option.

(A) Zoospore	(i) Endogenous
(B) Conidia	(ii) Exogenous
(C) Aplanospore	
(D) Ascospore	

- (1) (A)-(i), (B)-(ii), (C)-(i), (D)-(i)
- (2) (A)-(i), (B)-(ii), (C)-(i), (D)-(ii)
- (3) (A)-(i), (B)-(i), (C)-(ii), (D)-(i)
- (4) (A)-(ii), (B)-(i), (C)-(ii), (D)-(ii)

121. Which of the following disease is caused by viruses ?

- (1) Small pox
- (2) Potato spindle tuber disease
- (3) Mad cow disease in cattle
- (4) Cr-Jakob disease

122. Chlorophyll-bearing, simple, thalloid autotrophic and eukaryotic organisms are belong to:-

- (1) Fungi
- (2) Algae
- (3) Bacteria
- (4) Gymnosperm

123. How many of the followings are the members of chlorophyceae :-

Volvox, Fucus, Ectocarpus, Chara, Sargassum, Ulothrix, Gracilaria, Spirogyra

Choose the correct option :-

- (1) Five
- (2) Three
- (3) Seven
- (4) Four

124. Match column-I with column-II and select the correct option

Column-I		Column-II	
(A)	<i>Fucus</i>	I.	Homosporous pteridophyta
(B)	<i>Pteris</i>	II.	Brown algae
(C)	<i>Ginkgo</i>	III.	Heterosporous pteridophyta
(D)	<i>Salvinia</i>	IV.	Naked seed

- (1) (A) IV, (B) III, (C) II, (D) I
- (2) (A) II, (B) III, (C) I, (D) IV
- (3) (A) II, (B) IV, (C) I, (D) III
- (4) (A) II, (B) I, (C) IV, (D) III

125. How many of the followings are the members of Rhodophyceae :-

Polysiphonia, Porphyra, Fucus, Gracilaria, Gelidium, Laminaria, Dictyota.

Choose the correct option:-

- (1) Three
- (2) Two
- (3) Six
- (4) Four

126. Match the following and select correct option.

(A) Buds of protonema	(i) Haploid
(B) Antheridia	(ii) Diploid
(C) Archegonia	
(D) Spore mother cell	

- (1) (A)-(i), (B)-(i), (C)-(i), (D)-(ii)
- (2) (A)-(i), (B)-(ii), (C)-(ii), (D)-(ii)
- (3) (A)-(ii), (B)-(i), (C)-(i), (D)-(i)
- (4) (A)-(ii), (B)-(ii), (C)-(ii), (D)-(i)

127. Match List-I with List-II

	List-I		List-II
A	Gemma cup	i	<i>Sphagnum</i>
B	Peat Moss	ii	<i>Funaria</i> and <i>Polytrichum</i>
C	Mosses	iii	<i>Marchantia</i>
D	Capsule	iv	Part of sporophyte in bryophytes

Choose the correct answer from the option given below :

	A	B	C	D
1	iii	i	ii	iv
2	ii	iii	iv	i
3	ii	iii	i	iv
4	iii	iv	ii	i

128. Members of Pteridophyta -

- A. They produce spores by mitosis in spore mother cells.
- B. They form spores which germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.
- C. They perform fusion of male gamete with the megaspore present in the archegonium result in the formation of zygote.

Choose the correct option :-

- (1) Only A is incorrect.
- (2) Only B is incorrect.
- (3) Only C is incorrect.
- (4) Both A and C are incorrect.

129. Which of the following statement is incorrect ?

- (1) In Sphenopsida class, *equisetum* is included
- (2) Pteridophytes are used for medicinal purposes and as soil binders
- (3) Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous
- (4) *Salvinia* belongs to class lycopsida in pteridophytes

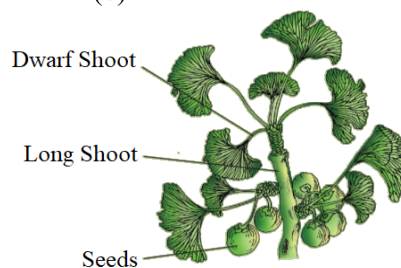
130. Identify the diagrams (a), (b) and (c) :-



(a)



(b)



(c)

- (1) (a) *Cycas*, (b) *Pinus*, (c) *Ginkgo*
- (2) (a) *Pinus*, (b) *Cycas*, (c) *Ginkgo*
- (3) (a) *Ginkgo*, (b) *Cycas*, (c) *Pinus*
- (4) (a) *Cycas*, (b) *Ginkgo*, (c) *Pinus*

131. If a root cell of *Cycas* plant contain 24 chromosome then what will be the number of chromosome in female gametophyte cell ?

- (1) 24 (2) 36 (3) 12 (4) 48

132. Higher gymnosperms have following character except:

- (1) Heterosporous nature
- (2) Wind-pollination
- (3) Nucellus is protected by envelopes
- (4) Gametophyte is monoecious

133. Sporophytic plant body of *Pinus* have :-

- (1) Male and female cone found on same plant body.
- (2) Unbranched stem.
- (3) Coralloid roots
- (4) All of the above

134. Match list-I and list-II :-

	List-I		List-II
(A)	Pteropsida	(i)	<i>Lycopodium</i>
(B)	Lycopsida	(ii)	<i>Psilotum</i>
(C)	Sphenopsida	(iii)	<i>Pteris</i>
(D)	Psilopsida	(iv)	<i>Equisetum</i>

Choose the correct answer from the option given below :-

- (1) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)
- (2) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)
- (3) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)
- (4) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

135. Gymnosperms are:

- (1) Covered seeded vascular embryophytes
- (2) Naked seeded non-vascular embryophytes
- (3) Naked seeded vascular non-embryophytes
- (4) Naked seeded vascular embryophytes

136. Tube within tube body plan is absent in :

- (1) *Taenia*
- (2) *Ascaris*
- (3) Earthworms
- (4) Cockroach

137. Which one is correct about Porifera :-

- (1) Unisexual
- (2) Internal fertilisation with indirect development
- (3) External fertilisation with direct development
- (4) External fertilisation with indirect development

138. Which of the following is exclusively marine :-

- (1) Sea walnut
- (2) *Amphioxus*
- (3) *Ascidia*
- (4) All above

139. A vertebrate which doesn't possess jaws :

- (1) *Balanoglossus*
- (2) *Petromyzon*
- (3) *Salamandra*
- (4) *Scoliodon*

140. Select the correct option :

Animal	Common name
(a) <i>Gorgonia</i>	(1) Bath sponge
(b) <i>Ctenoplana</i>	(2) Sea fan
(c) <i>Taenia</i>	(3) Comb jelly
(d) <i>Euspongia</i>	(4) Tapeworm

- (1) a-3, b-2, c-4, d-1
- (2) a-1, b-3, c-4, d-2
- (3) a-2, b-3, c-1, d-4
- (4) a-2, b-3, c-4, d-1

141. Metagenesis means :-

- (1) Polyp produce medusae asexually and medusae produce polyp sexually
- (2) Polyp produce medusae sexually and medusae produce polyp sexually
- (3) Polyp produce medusae asexually and medusae produce polyp asexually.
- (4) Both (1) & (2)

142. The animal group where the adults are less developed but larvae are more developed is :-

- (1) Agnatha
- (2) Tunicate
- (3) Amphibian
- (4) Cephalochordates

143. Animals are classified on the basis of :-

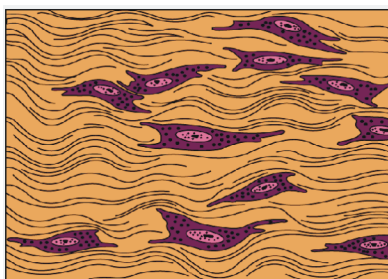
- (1) Arrangement of cells
- (2) Body symmetry
- (3) Nature of coelom
- (4) All of these

- 144.** In the given examples, how many animals are viviparous ?
Corvus, Columba, Macropus, Pteropus, Calotes, Hemidactylus, Psittacula, Struthio and Macaca
 (1) Three (2) Four (3) Five (4) Six
- 145.** Which of the following are incorrect for platyhelminthes ?
 (1) All are endoparasite
 (2) All are bilateral & Triploblastic
 (3) Hooks and suckers are present in the parasitic forms
 (4) Both 1 & 2
- 146.** Cyclostomes do not have :-
 (1) Scales (2) Caudal fins
 (3) Unpaired fins (4) Notochord
- 147.** Read the given statements and select the correct option.
Statement-1 : All triploblastic animals are eucoelomates.
Statement-2 : Diploblastic animals have a false coelom.
 (1) Both statements 1 and 2 are correct.
 (2) Statement 1 is not correct but statement 2 is correct.
 (3) Statement 1 is correct and statement 2 is not correct.
 (4) Both statements 1 and 2 are incorrect.
- 148.** How many of the following animals shows organ system level of body organisation along with bilateral symmetry ?
Pleurobrachia, Euspongia, Ancylostoma, Hirudinaria, Anopheles, Asterias, Balanoglossus
 (1) Four (2) Five
 (3) Two (4) Three
- 149.** Ciliated epithelial cells are known to occur in :-
 (1) Eustachian tube and stomach
 (2) Fallopian tube and bronchioles
 (3) Fallopian tube and urethra
 (4) Trachea and oesophagus.
- 150.** Cartilage is formed by :-
 (1) Osteoblasts (2) Fibroblasts
 (3) Chondrocytes (4) Osteocytes
- 151.** How many statements are **true** in the following?
 (i) Simple epithelium forms lining for body cavities, ducts and tubes.
 (ii) In epithelial tissue, cells are loosely packed with little intercellular matrix.
 (iii) Simple epithelium provide protection against chemical and mechanical stresses.
 (iv) Gap junction perform cementing to keep neighbouring cells together.
 (1) Two
 (2) Three
 (3) Four
 (4) One
- 152.** Which of the following connective tissue connects skeletal muscles to bones in form of tendon :-
 (1) Areolar connective tissue
 (2) Adipose tissue
 (3) Dense regular connective tissue
 (4) Dense irregular connective tissue
- 153.** PCT of nephron in kidney has :-
 (1) Simple squamous epithelium with microvilli
 (2) Simple cuboidal epithelium with microvilli
 (3) Simple columnar epithelium with microvilli
 (4) Simple epithelium with cilia and microvilli

154. What is the correct function of "tight junction" ?

- (1) Facilitate the cells to communicate with each other.
- (2) To stop substances from leaking across a tissue.
- (3) To perform cementing to keep neighbouring cells together.
- (4) To allow transport of substances in unidirection.

155.



The given figure represent :-

- (1) Adipose tissue (2) Areolar tissue
- (3) Cartilage (4) Ligament

156. Which type of cells of epithelium get specialised for secretion and are called glandular epithelium :-

- (a) Squamous (b) Cuboidal
- (c) Columnar

- (1) Only a (2) Only b
- (3) Only c (4) b and c

157. Function of compound epithelium is :-

- (1) Mainly Absorption
- (2) Only Secretion
- (3) Only Excretion
- (4) Mainly Covering & Protection

158. Highest regeneration power is present in :

- (1) Epithelial tissue
- (2) Connective tissue
- (3) Muscular tissue
- (4) Nervous tissue

159. For female cockroaches, which of the following statements are **correct**?

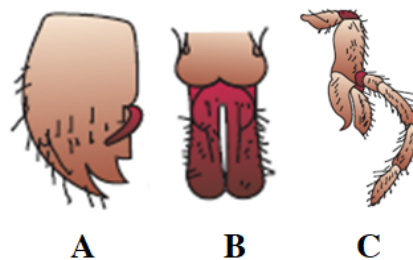
- (a) The 7th sternum is boat shaped
- (b) The 7th sternum together with the 8th and 9th sternum forms a brood or genital pouch
- (c) The anterior part of genital pouch contains female gonopore, spermathecal pores and collateral glands
- (d) Only right collateral gland form ootheca

- (1) a and b (2) b and d
- (3) a, b, c (4) a and d

160. Another name of Labrum is ?

- (1) Upper lip (2) Lower lip
- (3) Upper jaw (4) Lower jaw

161. Identify the structures from figures A, B, C which is not correctly matched with their function.



- (1) A – Mandible → Help in grinding.
- (2) B – Labrum → Upper lip of cockroach.
- (3) C – Maxilla → Form lateral wall of pre oral cavity.
- (4) B – Hypopharynx → Tongue

162. Which of the following is not a part of cockroach body associated with excretion ?

- (1) Malpighian tubules
- (2) Nephrocytes
- (3) Uricose gland of male reproductive system
- (4) Collateral glands of female reproductive system

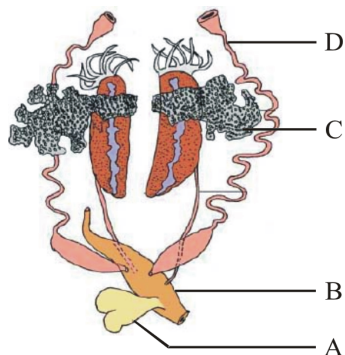
163. Fore wings of cockroach are :-

- (1) meta thoracic wings known as Tegmina
- (2) Pro thoracic wings known as Tegmina
- (3) Meso thoracic wings known as Tergites
- (4) Meso thoracic wings known as Tegmina

164. In male cockroaches, sperms are stored in which part of the reproductive system ?

- (1) Testes
- (2) Vas deferens
- (3) Seminal vesicles
- (4) Mushroom gland

165. Given below is a diagrammatic representation of female reproductive system of frog labelled A, B, C and D :



Find out incorrect one :-

- (1) There is functional connection between C and kidney.
- (2) B is small, median chamber that is used to pass faecal matter, urine and ova to the exterior.
- (3) D is one pair arising from the ovaries opens into the cloaca separately.
- (4) A is thin walled urinary bladder is present ventral to rectum which also opens in cloaca.

166. In frogs, testes are attached to kidneys by a double fold of peritoneum known as :-

- (1) Urinary bladder
- (2) Mesovarium
- (3) Urinogenital duct
- (4) Mesorchium

167. Sinus venosus in the heart of frog joins the

- (1) Left atrium
- (2) Ventricle
- (3) Truncus arteriosus
- (4) Right atrium

168. Which of the following is **incorrect** w.r.t Amphibians ?

- (1) Skull is dicondylid
- (2) They can respire through skin only
- (3) Some amphibians are found in sea
- (4) Both (2) and (3)

169. Which one is correct feature for given animal specimen :



- (1) 4 pair of gill slits
- (2) Sucking and circular mouth with Jaw
- (3) Marine and show migration
- (4) All of these

170. Limbless amphibian is :

- (1) Hyla
- (2) Ichthyophis
- (3) Salamandra
- (4) Bufo

171. Select the incorrect statement for chordates :

- (1) They possess a post anal tail and a closed circulatory system.
- (2) Protochordata subphylum include class urochordata and cephalochordata.
- (3) Heart is ventral.
- (4) Single dorsal hollow nerve cord present.

172. Match the column I & II and select the correct option :

Column-I		Column-II	
(i)	Myxine	(a)	Mammalia
(ii)	Aptenodyte	(b)	Cyclostomata
(iii)	Felis	(c)	Urochordata
(iv)	Salpa	(d)	Aves

- (1) (i)-a, (ii)-b, (iii)-c, (iv)-d
- (2) (i)-b, (ii)-d, (iii)-a, (iv)-c
- (3) (i)-d, (ii)-c, (iii)-a, (iv)-b
- (4) (i)-d, (ii)-c, (iii)-b, (iv)-a

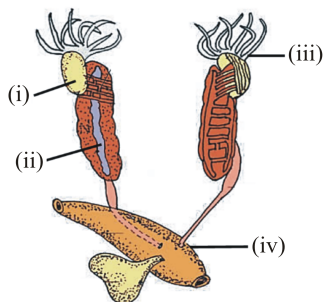
173. Assertion (A) : Main functions of simple cuboidal epithelium are secretion and absorption.
Reason (R) : The function of ciliated epithelium is to move particles in a specific direction.

- (1) Both (A) and (R) is correct and (R) is correct explanation of (A).
- (2) Both (A) and (R) is correct but (R) is not correct explanation of (A).
- (3) Both (A) and (R) incorrect
- (4) (A) is correct but (R) is incorrect.

174. Most of the exchange surfaces of multicellular animals are lined with :

- (1) Epithelial tissue
- (2) Connective tissue
- (3) Neural tissue
- (4) Smooth muscle cells

175. Given below is the diagram of reproductive system of an animal labelled by (i), (ii), (iii) and (iv), identify the animal with correct labelling of (i) to (iv)



- (1) Male frog-(i) Testes (ii) Adrenal gland (iii) Fat bodies (iv) Cloaca
- (2) Female frog-(i) Kidney (ii) Adrenal gland (iii) Ovary (iv) Cloaca
- (3) Male frog-(i) Testes (ii) Bidder canal (iii) Adrenal gland (iv) Cloaca
- (4) Female frog-(i) Ovary (ii) Oviduct (iii) Adrenal gland (iv) Cloaca

176. Vasa-efferentia in male frog open into :

- (1) Vas deferens
- (2) Rete testis
- (3) Cloaca
- (4) Bidder's canal

177. Frog can protect it self from enemies by ?

- (1) Teeth
- (2) Camouflage
- (3) Clawed digit
- (4) Both (1) and (2)

178. In frog ureter act as urinogenital duct in :-

- (1) Only in male
- (2) Only in female
- (3) Both male & female
- (4) It is absent in frog

179. The number of ganglia in thorax and abdomen of cockroach are respectively.

- (1) Three and Six
- (2) Six and Three
- (3) Two and Eight
- (4) Three and Seven

180. Arthrodial membrane (Articular membrane) connects the :

- (1) Legs of cockroach
- (2) Antennal segments of cockroach
- (3) Sclerites of cockroach
- (4) Head to thorax of cockroach

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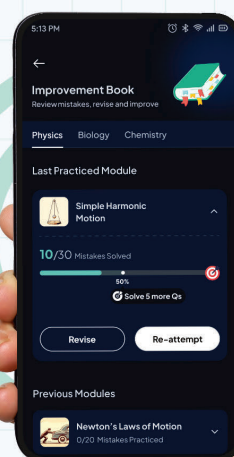
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1	13-03-2025	THURSDAY	SYLLABUS -1
2	17-03-2025	MONDAY	SYLLABUS -2
3	21-03-2025	FRIDAY	SYLLABUS -3
4	25-03-2025	TUESDAY	SYLLABUS -4
5	29-03-2025	SATURDAY	SYLLABUS -5
6	02-04-2025	WEDNESDAY	SYLLABUS -6
7	06-04-2025	SUNDAY	FULL SYLLABUS (A.I.O.T)
8	08-04-2025	TUESDAY	SYLLABUS -1 + 2 + 3
9	11-04-2025	FRIDAY	SYLLABUS -4 + 5 + 6
10	14-04-2025	MONDAY	FULL SYLLABUS
11	17-04-2025	THURSDAY	FULL SYLLABUS
12	20-04-2025	SUNDAY	FULL SYLLABUS (A.I.O.T)
13	22-04-2025	TUESDAY	FULL SYLLABUS
14	25-04-2025	FRIDAY	FULL SYLLABUS
15	28-04-2025	MONDAY	FULL SYLLABUS
16	01-05-2025	THURSDAY	FULL SYLLABUS
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SYLLABUS 2	» Center Of Mass And Collision » Circular Motion » Rotational Motion » Gravitation » Properties Of Solids And Liquids » Experimental Skills * Young'S Modulus Of Elasticity Of The Material Of A Metallic Wire. * Surface Tension Of Water By Capillary Rise And Effect Of Detergents. * Co-Efficient Of Viscosity Of A Given Viscous Liquid By Measuring Terminal Velocity Of A Given Spherical Body. * Metre Scale - The Mass Of A Given Object By The Principle Of Moments.	Atomic structure, Classification of elements and periodicity in properties, Chemical bonding & Molecular structure	Cell : The unit of life, Cell cycle and cell division, Biomolecules, Enzyme, Excretory products and their elimination (Excretory System) Breathing and exchange of gases (Respiratory System) Body fluids and circulation (Circulatory System)
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